

Manufacturing accounts

Learning objectives

After you have studied this chapter, you should be able to:

- calculate prime cost and production cost of goods manufactured
- draw up a manufacturing account and statement of profit or loss
- adjust the manufacturing account in respect of work-in-progress

Introduction

In this chapter, you'll learn how to prepare manufacturing accounts and the reasons for doing so.

37.1 Manufacturing: not retailing

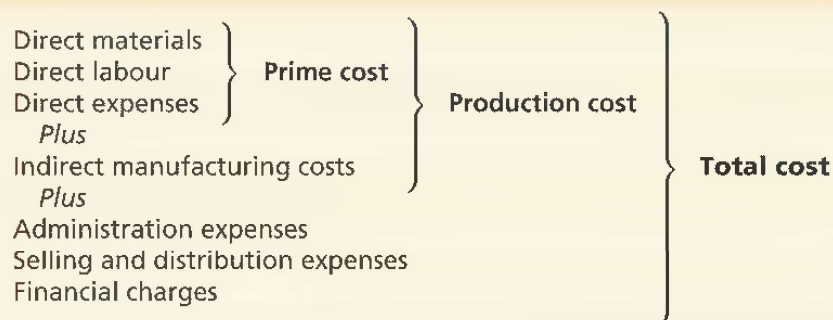
We now have to deal with businesses which are manufacturers. For these businesses, a **manufacturing account** is prepared in addition to the statement of profit or loss. **It is produced for internal use only.** People other than the owners and managers of the organisation concerned rarely see a manufacturing account.

If a business is using manufacturing accounts, instead of a figure for purchases (of finished goods) the trading account will contain the cost of manufacturing the goods that were manufactured during the period. The manufacturing account is used to calculate and show the cost of manufacturing those goods. The figure it produces that is used in the trading account is known as the **production cost**.

37.2 Divisions of costs

In a manufacturing business the costs are divided into different types. These may be summarised in chart form as in Exhibit 37.1:

Exhibit 37.1



The prime cost items and the other production cost items are shown in the manufacturing account. The administration expenses, selling and distribution expenses and the financial charges appear in the statement of profit or loss.

37.3 Direct and indirect costs

With reference to Exhibit 37.1, when you see the word *direct* followed by a type of cost, you know that it has been possible to trace the costs to an item being manufactured.

As shown in the chart, the sum of all the **direct costs** is known as the **prime cost**. If a manufacturing-related cost cannot easily be traced to the item being manufactured, then it is an indirect cost and will be included under **indirect manufacturing costs** (which are also sometimes known as 'factory overhead expenses'). 'Production cost' is the sum of prime cost plus the indirect manufacturing costs.

For example, the wages of a machine operator making a particular item will be direct labour. The wages of a foreman in charge of many men on different jobs will be indirect labour, and will be part of the indirect manufacturing costs. Other examples of costs being direct costs would be:

- 1 Cost of raw materials including carriage inwards on those raw materials.
- 2 Hire of special machinery for a job.

Activity 37.1

Think about it for a minute and then list five costs you think are direct and five that you think are indirect.

37.4 Indirect manufacturing costs

'Indirect manufacturing costs' are all those costs which occur in the factory or other place where production is being done, but which cannot easily be traced to the items being manufactured. Examples are:

- wages of cleaners
- wages of crane drivers
- rent of a factory
- depreciation of plant and machinery
- costs of operating forklift trucks
- factory power
- factory lighting.

37.5 Administration expenses

'Administration expenses' consist of such items as managers' salaries, legal and accountancy charges, the depreciation of accounting machinery and secretarial salaries.

37.6 Selling and distribution expenses

'Selling and distribution expenses' are items such as sales staff's salaries and commission, carriage outwards, depreciation of delivery vans, advertising and display expenses.

37.7 Financial charges

'Financial charges' are expense items such as bank charges and discounts allowed.

Activity 37.2

Place a tick in the appropriate column for each of the following cost items:

	<i>Direct materials</i>	<i>Direct labour</i>	<i>Direct expenses</i>	<i>Indirect manufacturing costs</i>	<i>Administration expenses</i>	<i>Selling and distribution expenses</i>	<i>Financial charges</i>
(a) Purchases of raw materials							
(b) Direct wages							
(c) General factory expenses							
(d) Depreciation of machinery							
(e) Commission on sales							
(f) Factory rent							
(g) Carriage inwards of raw materials							
(h) Royalties paid							
(i) Inventory of raw materials							
(j) Administration salaries							
(k) Indirect labour							
(l) Bank charges							
(m) Carriage outwards							
(n) Discounts allowed							
(o) Factory lighting							

37.8 Format of financial statements

Manufacturing account section

This is debited with the production cost of goods completed during the accounting period. It contains costs of:

- direct materials;
- direct labour;
- direct expenses; and
- indirect manufacturing costs.

The manufacturing account includes all purchases of raw materials, including the inventory adjustments for raw materials. It also includes inventory adjustments for **work-in-progress** (goods that are partly completed at the end of a period). Let's put this into a series of steps:

- 1 Add opening inventory of raw materials to purchases and subtract the closing inventory of raw materials.
- 2 Add in all the direct costs to get the prime cost.
- 3 Add in all the indirect manufacturing costs.
- 4 Add the opening inventory of work-in-progress and subtract the closing inventory of work-in-progress to get the production cost of all goods completed in the period.

Thus, when completed, the manufacturing account shows the total of production cost that relates to those manufactured goods that have been available for sale during the period. This figure will then be transferred down to the statement of profit or loss where it will replace the entry for purchases.

Trading account section of the statement of profit or loss

This account includes:

- production cost brought down from the manufacturing account
- opening and closing inventory of finished goods
- sales.

When completed, this account shows the gross profit. This is then carried down to the profit and loss account part.

The manufacturing account and the trading account can be shown as in Exhibit 37.2.

Exhibit 37.2

Manufacturing Account			
Production costs for the period:			£
Direct materials			xxx
Direct labour			xxx
Direct expenses			<u>xxx</u>
Prime cost			xxx
Indirect manufacturing costs			xxx
Production cost of goods completed c/d to trading account			<u>xxx</u>
Trading Account			
		£	£
Sales			xxx
Less Production cost of goods sold:			
Opening inventory of finished goods	(A)	xxx	
→ Add Production costs of goods completed b/d		<u>xxx</u>	
		xxx	
Less Closing inventory of finished goods	(B)	<u>(xxx)</u>	
Gross profit			(xxx)
			<u>xxx</u>

(A) is production costs of goods unsold in previous period.

(B) is production costs of goods unsold at end of the current period.

Profit and loss section of the statement of profit or loss

This is prepared in the way you learnt in earlier chapters in this book. You know, therefore, that it includes:

- gross profit brought down from the trading account
- all administration expenses
- all selling and distribution expenses
- all financial charges.

However, some of the items you would normally put in the profit and loss account part are already included in the manufacturing account, e.g. depreciation on machines, and canteen wages. When completed, this account will show the net profit.

**Activity
37.3**

Why do you think some expenses have been moved to the manufacturing account?

37.9 A worked example of a manufacturing account

Exhibit 37.3 shows the necessary details for a manufacturing account. It has been assumed that there were no partly completed units (work-in-progress) either at the beginning or end of the period.

Exhibit 37.3

Details of production costs for the year ended 31 December 2014:

	£
1 January 2014, inventory of raw materials	5,000
31 December 2014, inventory of raw materials	7,000
Raw materials purchased	80,000
Manufacturing (direct) wages	210,000
Royalties	1,500
Indirect wages	90,000
Rent of factory – excluding administration and selling and distribution blocks	4,400
Depreciation of plant and machinery in factory	4,000
General indirect expenses	3,100

Manufacturing Account for the year ending 31 December 2014

	£	£
Inventory of raw materials 1.1.2014		5,000
Add Purchases		<u>80,000</u>
		85,000
Less Inventory of raw materials 31.12.2014		<u>(7,000)</u>
Cost of raw materials consumed		78,000
Manufacturing wages		210,000
Royalties		<u>1,500</u>
Prime cost		289,500
Indirect manufacturing costs		
Rent	4,400	
Indirect wages	90,000	
General expenses	3,100	
Depreciation of plant and machinery	<u>4,000</u>	
		<u>101,500</u>
Production cost of goods completed c/d		<u>391,000</u>

Sometimes, if a business has produced less than the customers have demanded, it may buy in some finished goods. In this case, the trading account will have both a figure for purchases of finished goods and a figure for production cost of goods completed.

37.10 Work-in-progress

The production cost to be carried down to the trading account is that of production cost of goods completed during the period. If items have not been completed, they cannot be sold. Therefore, they should not appear in the trading account.

For instance, if we have the following information, we can calculate the transfer to the trading account:

	£
Total production costs expended during the year	50,000
Production costs last year on goods not completed last year, but completed in this year (work-in-progress)	3,000
Production costs this year on goods which were not completed by the year end (work-in-progress)	4,400

The calculation is:

Total production costs expended this year	50,000
Add Costs from last year, in respect of goods completed in this year (work-in-progress)	<u>3,000</u>
	53,000
Less Costs in this year, for goods to be completed next year (work-in-progress)	<u>(4,400)</u>
Production costs expended on goods completed this year	<u>48,600</u>

37.11 Another worked example

Exhibit 37.4

	£
1 January 2014, Inventory of raw materials	8,000
31 December 2014, Inventory of raw materials	10,500
1 January 2014, Work-in-progress	3,500
31 December 2014, Work-in-progress	4,200
Year to 31 December 2014:	
Wages: Direct	39,600
Indirect	25,500
Purchase of raw materials	87,000
Fuel and power	9,900
Direct expenses	1,400
Lubricants	3,000
Carriage inwards on raw materials	2,000
Rent of factory	7,200
Depreciation of factory plant and machinery	4,200
Internal transport expenses	1,800
Insurance of factory buildings and plant	1,500
General factory expenses	3,300

This information produces the following manufacturing account:

Manufacturing Account for the year ending 31 December 2014

	£	£
Inventory of raw materials 1.1.2014		8,000
Add Purchases		87,000
Carriage inwards		<u>2,000</u>
		97,000
Less Inventory of raw materials 31.12.2014		<u>(10,500)</u>
Cost of raw materials consumed		86,500
Direct wages		39,600
Direct expenses		<u>1,400</u>
Prime cost		127,500
Indirect manufacturing costs:		
Fuel and power	9,900	
Indirect wages	25,500	
Lubricants	3,000	
Rent	7,200	
Depreciation of plant and machinery	4,200	
Internal transport expenses	1,800	
Insurance	1,500	
General factory expenses	<u>3,300</u>	
		56,400
		183,900
Add Work-in-progress 1.1.2014		<u>3,500</u>
		187,400
Less Work-in-progress 31.12.2014		<u>(4,200)</u>
Production cost of goods completed c/d		<u>183,200</u>

The trading account is concerned with finished goods. If in the above example there had been £3,500 inventory of finished goods at 1 January 2014 and £4,400 at 31 December 2014, and the sales of finished goods amounted to £250,000 then the trading account would be:

Trading Account for the year ending 31 December 2014

	£	£
Sales		250,000
Less Cost of goods sold:		
Inventory of finished goods 1.1.2014	3,500	
Add Production cost of goods completed b/d	<u>183,200</u>	
	186,700	
Less Inventory of finished goods 31.12.2014	<u>(4,400)</u>	
		182,300
Gross profit c/d		<u><u>67,700</u></u>

The profit and loss section is then constructed in the normal way.

37.12 Apportionment of expenses

Quite often expenses will have to be split between –

- Indirect manufacturing costs: to be charged in the manufacturing account section; and
- Administration expenses:
- Selling and distribution expenses: } to be charged in the profit and loss section.
- Financial charges:

An example of this could be the rent expense. If the rent is paid separately for each part of the organisation, then it is easy to charge the rent to each sort of expense. However, only one figure of rent may be paid, without any indication as to how much is for the factory, how much is for the selling and distribution building and how much is for the administration building.

How the rent expense will be apportioned in the latter case will depend on the circumstances, using the most equitable way of doing it. A range of methods may be used. Common ones include apportionment on the basis of:

- floor area
- property valuations of each part of the buildings and land.

37.13 Full set of financial statements

A complete worked example is now given. Note that in the profit and loss account part the expenses have been separated so as to show whether they are administration expenses, selling and distribution expenses, or financial charges.

The trial balance in Exhibit 37.5 has been extracted from the books of J. Jarvis, Toy Manufacturer, as at 31 December 2014.

Exhibit 37.5

J. Jarvis			
Trial Balance as at 31 December 2014			
	Dr	Cr	
	£	£	
Inventory of raw materials 1.1.2014	21,000		
Inventory of finished goods 1.1.2014	38,900		
Work-in-progress 1.1.2014	13,500		
Wages (direct £180,000; factory indirect £145,000)	325,000		
Royalties	7,000		
Carriage inwards (on raw materials)	3,500		
Purchases of raw materials	370,000		
Productive machinery (cost £280,000)	230,000		
Administration computers (cost £20,000)	12,000		
General factory expenses	31,000		
Lighting	7,500		
Factory power	13,700		
Administration salaries	44,000		
Sales reps' salaries	30,000		
Commission on sales	11,500		
Rent	12,000		
Insurance	4,200		
General administration expenses	13,400		
Bank charges	2,300		
Discounts allowed	4,800		
Carriage outwards	5,900		
Sales			1,000,000
Accounts receivable and accounts payable	142,300		64,000
Bank	16,800		
Cash	1,500		
Drawings	60,000		
Capital as at 1.1.2014			357,800
	<u>1,421,800</u>		<u>1,421,800</u>

Notes at 31.12.2014:

- 1 Inventory of raw materials £24,000; inventory of finished goods £40,000; work-in-progress £15,000.
- 2 Lighting, rent and insurance are to be apportioned: factory $\frac{5}{6}$, administration $\frac{1}{6}$.
- 3 Depreciation on productive machinery and administration computers at 10 per cent per annum on cost.

J. Jarvis

Manufacturing Account and Statement of Profit or Loss for the year ending 31 December 2014

	£	£	£
Inventory of raw materials 1.1.2014			21,000
Add Purchases			370,000
Carriage inwards			3,500
			<u>394,500</u>
Less Inventory raw materials 31.12.2014			(24,000)
Cost of raw materials consumed			<u>370,500</u>
Direct labour			180,000
Royalties			7,000
Prime cost			<u>557,500</u>
Indirect manufacturing costs:			
General factory expenses	31,000		
Lighting $\frac{5}{6}$	6,250		
Power	13,700		
Rent $\frac{5}{6}$	10,000		
Insurance $\frac{5}{6}$	3,500		
Depreciation of productive machinery	28,000		
Indirect labour	<u>145,000</u>		
			<u>237,450</u>
			<u>794,950</u>
Add Work-in-progress 1.1.2014			13,500
			<u>808,450</u>
Less Work-in-progress 31.12.2014			(15,000)
Production cost of goods completed c/d			<u>793,450</u>
Sales			1,000,000
Less Cost of goods sold:			
Inventory of finished goods 1.1.2014	38,900		
Add Production cost of goods completed b/d	<u>793,450</u>		
			<u>832,350</u>
Less Inventory of finished goods 31.12.2014			(40,000)
			<u>(792,350)</u>
Gross profit			<u>207,650</u>
Administration expenses			
Administration salaries	44,000		
Rent $\frac{1}{6}$	2,000		
Insurance $\frac{1}{6}$	700		
General expenses	13,400		
Lighting $\frac{1}{6}$	1,250		
Depreciation of administration computers	<u>2,000</u>		
		63,350	
Selling and distribution expenses			
Sales reps' salaries	30,000		
Commission on sales	11,500		
Carriage outwards	<u>5,900</u>		
		47,400	
Financial charges			
Bank charges	2,300		
Discounts allowed	<u>4,800</u>		
		<u>7,100</u>	
Net profit			<u>(117,850)</u>
			<u>89,800</u>



J. Jarvis
Statement of Financial Position as at 31 December 2014

	£	£
<i>Non-current assets</i>		
Productive machinery at cost	280,000	
Less Depreciation to date	<u>(78,000)</u>	
		202,000
Administration computers at cost	20,000	
Less Depreciation to date	<u>(10,000)</u>	
		<u>10,000</u>
		212,000
<i>Current assets</i>		
Inventory		
Raw materials	24,000	
Finished goods	40,000	
Work-in-progress	15,000	
Accounts receivable	142,300	
Bank	16,800	
Cash	<u>1,500</u>	
		239,600
		451,600
<i>Less Current liabilities</i>		
Accounts payable		<u>(64,000)</u>
		<u>387,600</u>
<i>Financed by</i>		
<i>Capital</i>		
Balance as at 1.1.2014		357,800
Add Net profit		<u>89,800</u>
		447,600
Less Drawings		<u>(60,000)</u>
		<u>387,600</u>

37.14 Market value of goods manufactured

The financial statements of Jarvis, just illustrated, are subject to the limitation that the respective amounts of the gross profit which are attributable to the manufacturing side or to the selling side of the business are not known. A technique is sometimes used to bring out this additional information. This method uses the cost which would have been involved if the goods had been bought in their finished state instead of being manufactured by the business. This figure is credited to the manufacturing account and debited to the trading account so as to throw up two figures of gross profit instead of one. It should be pointed out that the net profit will remain unaffected. All that will have happened will be that the figure of £207,650 gross profit will be shown as two figures instead of one. When added together, they will total £207,650.

Assume that the cost of buying the goods instead of manufacturing them had been £950,000. The relevant parts of the Manufacturing Account and Statement of Profit or Loss will then be:

**Manufacturing Account and Statement of Profit or Loss extract for the year ending
31 December 2014**

	£	£
Market value of goods completed		950,000
Less Production cost of goods completed (as before)		<u>(793,450)</u>
Gross profit on manufacture c/d		<u>156,550</u>
Sales		1,000,000
Inventory of finished goods 1.1.2014	38,900	
Add Market value of goods completed b/d	<u>950,000</u>	
	988,900	
Less Inventory of finished goods 31.12.2014	<u>(40,000)</u>	
		<u>(948,900)</u>
Gross profit on trading c/d		51,100
Gross profit		
On manufacturing	<u>156,550</u>	
On trading	<u>51,100</u>	
		<u>207,650</u>

Learning outcomes

You should now have learnt:

- 1 Why manufacturing accounts are used.
- 2 How to prepare a manufacturing account and statement of profit or loss.
- 3 That the trading account section of the statement of profit or loss is used for calculating the gross profit made by selling the goods manufactured.
- 4 That the profit and loss account section of the statement of profit or loss shows as net profit what is left of gross profit after all administration, selling and distribution and finance costs incurred have been deducted.
- 5 That work-in-progress, both at the start and the close of a period, must be adjusted so as to identify the production costs of goods completed in the period.

Answers to activities

37.1 You may have included some of the following:

<i>Direct costs</i>	<i>Indirect costs</i>
(1) raw materials	canteen wages
(2) machine operator's wages	business rates
(3) packer's wages	rent
(4) machine set-up costs	insurance
(5) crane hire for building contract	storage of finished goods costs

However, you can only really do a split like this if you have a specific job or product in mind. You must first identify the 'cost object', that is, the item you are making or providing. Taking the example of a construction company building a hotel (it is engaged in other similar projects at the same time). The direct and indirect costs may include:

<i>Direct costs</i>	<i>Indirect costs</i>
(1) concrete	site canteen wages
(2) forklift truck operator's wages	company lawyer's salary
(3) bricklayer's wages	company architect's salary
(4) steel girders	company headquarters insurance
(5) windows	company warehousing costs

Now you should see that the indirect costs are not solely incurred in order to build the hotel. This is the key. Direct costs are those costs you can specifically link to a specific job. All the other costs of a job are indirect.

- 37.2** Direct materials (a) (g) (i)
 Direct labour (b)
 Direct expenses (h)
 Indirect manufacturing costs (c) (d) (f) (k) (o)
 Administration expenses (j)
 Selling and distribution expenses (e) (m)
 Financial charges (l) (n)

- 37.3** Because only administration expenses, selling and distribution expenses, and financial charges appear in the profit and loss account part when a manufacturing account is being used. The rest all arose because manufacturing was taking place and can be directly or indirectly attributed to the products being produced, so they appear in the manufacturing account.

Review questions

37.1 A business both buys loose tools and also makes some itself. The following data is available concerning the years ended 31 December 2014, 2015 and 2016.

		£
2014		
Jan 1	Inventory of loose tools	6,000
	During the year:	
	Bought loose tools from suppliers	8,000
	Made own loose tools: the cost of wages of employees being £1,300 and the materials cost £900	
Dec 31	Loose tools valued at	12,000
2015		
	During the year:	
	Loose tools bought from suppliers	4,000
	Made own loose tools: the cost of wages of employees being £1,600 and the materials cost £1,200	
Dec 31	Loose tools valued at	12,800
2016		
	During the year:	
	Loose tools bought from suppliers	7,200
	Made own loose tools: the cost of wages of employees being £1,000 and the materials cost £1,340. Received refund from a supplier for faulty tools returned to him	320
Dec 31	Loose tools valued at	14,600

You are to draw up the Loose Tools Account for the three years, showing the amount transferred as an expense in each year to the Manufacturing Account.

37.2 Using whichever of the following figures are required, prepare a manufacturing account and trading account for 2017. The manufacturing account should show clearly the prime cost of manufacture and the production cost of finished goods produced.

	£
Inventory, 1 January 2017:	
Raw materials	24,600
Partly finished goods	20,200
Finished goods	18,600
Inventory, 31 December 2017:	
Raw materials	31,300
Partly finished goods	23,400
Finished goods	29,200
Purchases of raw materials	165,400
Carriage on raw materials	9,100
Salaries and wages: factory (including £44,500 for management and supervision)	151,400
Salaries and wages: general office	28,400
Rent and business rates (three-quarters works, one-quarter office)	3,000
Lighting and heating (seven-eighths works, one-eighth office)	5,600
Repairs to machinery	3,400
Depreciation of machinery	5,600
Factory direct expenses	730
Insurance of plant and machinery	860
Sales	406,120

Note: partly finished goods are valued at their production cost.

37.3A From the following information, prepare a manufacturing account and statement of profit or loss for the year ending 31 December 2016 and a statement of financial position as at 31 December 2016 for J. Jones Limited.

	£	£
Purchase of raw materials	258,000	
Fuel and light	21,000	
Administration salaries	17,000	
Factory wages	59,000	
Carriage outwards	4,000	
Rent and business rates	21,000	
Sales		482,000
Returns inward	7,000	
General office expenses	9,000	
Repairs to plant and machinery	9,000	
Inventory at 1 January 2016:		
Raw materials	21,000	
Work-in-progress	14,000	
Finished goods	23,000	
Sundry accounts payable		37,000
Capital account		457,000
Freehold premises	410,000	
Plant and machinery	80,000	
Accounts receivable	20,000	
Accumulated provision for depreciation on plant and machinery		8,000
Cost in hand	<u>11,000</u>	<u> </u>
	<u>984,000</u>	<u>984,000</u>

Make provision for the following:

- (i) Inventory in hand at 31 December 2016:
 Raw materials £25,000
 Work-in-progress £11,000
 Finished goods £26,000.



-
- (ii) Depreciation of 10% on plant and machinery using the straight line method.
 - (iii) 80% of fuel and light and 75% of rent and rates to be charged to manufacturing.
 - (iv) Allowance for doubtful debts: 5% of sundry accounts receivable.
 - (v) £4,000 outstanding for fuel and light.
 - (vi) Rent and business rates paid in advance: £5,000.
 - (vii) Market value of finished goods: £382,000.

37.4 Prepare a manufacturing account and statement of profit or loss from the following balances of Z. Varga for the year ending 31 December 2017.

	£
Inventory at 1 January 2017:	
Raw materials	50,800
Work-in-progress	62,200
Finished goods	46,520
Purchases: Raw materials	183,070
Carriage on raw materials	3,920
Direct labour	168,416
Office salaries	66,838
Rent	10,400
Office lighting and heating	8,840
Depreciation: Works machinery	20,400
Office equipment	4,600
Sales	637,244
Factory fuel and power	16,240

Rent is to be apportioned: Factory $\frac{3}{4}$; Office $\frac{1}{4}$. Inventory at 31 December 2017 was: Raw materials £57,800; Work-in-progress £49,200; Finished goods £57,692.

37.5 From the following information, draw up a manufacturing account and the trading account section of the statement of profit or loss for the six months ending 30 September 2017. You should show clearly:

- (a) Cost of raw materials consumed.
- (b) Prime cost of production.
- (c) Production cost of finished goods.
- (d) Gross profit on sales.

	£
Inventory, 1 April 2017:	
Raw materials	2,990
Work-in-progress	3,900
Finished goods	15,300
Inventory, 30 September 2017:	
Raw materials	4,200
Work-in-progress	3,600
Finished goods	17,700
Purchases of raw materials	15,630
Carriage on raw materials	126
Direct wages	48,648
Factory general expenses	7,048
Office salaries	22,200
Depreciation of office furniture	420
Carriage outwards	191
Advertising	1,472
Bad debts	200
Sales less returns	112,410
Sales of scrap	1,317
Discounts received	188
Depreciation of factory equipment	4,200
Rent and business rates (factory three-quarters, office one-quarter)	2,800

37.6A From the following figures prepare a manufacturing account and the trading account section of the statement of profit or loss so as to show:

- (a) Cost of raw materials used in production.
- (b) Prime cost.
- (c) Production cost of finished goods produced.
- (d) Cost of goods sold.
- (e) Gross profit.

	£
Inventory at 1 January 2017:	
Raw materials	10,500
Goods in course of manufacture (at factory cost)	2,400
Finished goods	14,300
Inventory at 31 March 2017:	
Raw materials	10,200
Goods in course of manufacture (at factory cost)	2,900
Finished goods	13,200
Expenditure during the quarter:	
Purchases of raw materials	27,200
Factory wages: direct	72,600
indirect	13,900
Carriage on purchases of raw materials	700
Rent and business rates of the factory	1,200
Power	2,000
Depreciation of machinery	3,900
Repairs to factory buildings	1,300
Sundry factory expenses	900
Sales during the quarter	160,400

37.7 E. Wilson is a manufacturer. His trial balance at 31 December 2017 is as follows:

	£	£
Delivery van expenses	1,760	
Lighting and heating: Factory	7,220	
Office	1,490	
Manufacturing wages	72,100	
General expenses: Factory	8,100	
Office	1,940	
Sales reps: commission	11,688	
Purchase of raw materials	57,210	
Rent: Factory	6,100	
Office	2,700	
Machinery (cost £40,000)	28,600	
Office equipment (cost £9,000)	8,200	
Office salaries	17,740	
Accounts receivable	34,200	
Accounts payable		9,400
Bank	16,142	
Sales		194,800
Van (cost £6,800)	6,200	
Inventory at 31 December 2016:		
Raw materials	13,260	
Finished goods	41,300	
Drawings	24,200	
Capital		155,950
	<u>360,150</u>	<u>360,150</u>





Prepare the manufacturing account and statement of profit or loss for the year ending 31 December 2017 and a statement of financial position as at that date. Give effect to the following adjustments:

- 1 Inventory at 31 December 2017: raw materials £14,510; finished goods £44,490. There is no work-in-progress.
- 2 Depreciate machinery £3,000; office equipment £600; van £1,200.
- 3 Manufacturing wages due but unpaid at 31 December 2017 £550; office rent prepaid £140.

37.8 The financial year end of Mendip Limited is 30 June. At 30 June 2017, the following balances are available:

	£
Freehold land and buildings at cost	143,000
Plant and machinery at cost	105,000
Accumulated depreciation on plant and machinery	23,000
Purchase of raw materials	130,100
Sales	317,500
Factory rates	3,000
Factory heat and light	6,500
Accounts receivable	37,200
Accounts payable	30,900
Wages (including £15,700 for supervision)	63,000
Direct factory expenses	9,100
Selling expenses	11,000
Office salaries and general expenses	43,000
Bank	24,500
General reserve	30,000
Retained profits	18,000
Inventory 1 July 2016: Raw materials	20,000
Finished goods	38,000
Dividends paid: Preference shares	840
Ordinary shares	20,000

- (i) The inventory at 30 June 2017 was: raw materials £22,000; finished goods £35,600.
- (ii) Salaries include £6,700 for directors' fees.
- (iii) Depreciation is to be charged at 10% on cost of plant and machinery.

Required:

Prepare a manufacturing account and statement of profit or loss for the year ending 30 June 2017.